Minhao Zhang

Personal Data

Current position:	M.S. student, Center for Combinatorics, Nankai University.
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Research Interests

My research interests lie in algebraic combinatorics, especially symmetric functions and their combinatorics.

One central direction of my research concerns the shuffle theorem and its generalizations, which connect the Frobenius characteristic of certain S_n-modules with combinatorial models such as parking functions and related statistics. In particular, I am interested in the relationship between expressions of the form ∇f, where ∇ is the Bergeron–Garsia operator acting diagonally on modified Macdonald polynomials, and corresponding combinatorial objects. Such relationships can lead to Schur positivity, e-positivity, and other positivity phenomena.

Another direction I am interested in is the theory of nonsymmetric Macdonald polynomials. More specifically, I am interested in the interaction among nonsymmetric Macdonald theory, the affine Hecke algebra, and combinatorial models, as well as in possible nonsymmetric analogues of shuffle-theorem-type formulas.

Education

Nankai University, Center for Combinatorics — Sept. 2024–June 2027
M.S. in Applied Mathematics — GPA: 4.00/4.00
Advisor: Prof. Dun Qiu.
Thesis: TBD.

Nanjing Normal University, Honors College — Sept. 2020–June 2024
B.S. in Mathematics and Applied Mathematics — GPA: 3.85/4.00
Thesis: *A note on the number of the complete bipartite subgraphs.*
Thesis advisor: Prof. Alina F.Y. Zhao.

Papers

The extended (km, kn) shuffle theorem, with Dun Qiu. — 2026
Manuscript in preparation

- We extend the (km, kn) Shuffle Theorem to partially labeled rational parking functions by allowing zero labels. The algebraic side is expressed in terms of Negut elements through the Catalanimal framework.
- We also establish an extended Delta-type refinement in the (km, kn) setting.

The Schur positivity of ∇m_μ, with Dun Qiu. — 2026
arXiv:2607.00940

- Proves the signed Schur positivity of ∇m_μ, conjectured by Bergeron, Garsia, Haiman, and Tesler in 1999, and extends the result to $\nabla^r m_\mu$.
- We expand m_μ in terms of $C_\alpha(1)$, where the C_α are operators appearing in the compositional Shuffle Theorem. We also establish an e-positive analogue.

The nonsymmetric compositional Delta theorem, with Dun Qiu. — 2026
arXiv:2604.10226

- Extends the compositional Delta theorem to a nonsymmetric setting, using flagged LLT polynomials to formulate signed and unsigned nonsymmetric identities.
- Introduces nonsymmetric variants of the ∇ and τ^* operators and shows that Weyl symmetrization recovers the classical compositional Delta theorem.

Teaching Experience

Nankai University	2024–present
Teaching Assistant, Linear Algebra (undergraduate course)	Fall 2026
Teaching Assistant, Calculus I (undergraduate course)	Fall 2026
Teaching Assistant, Macdonald Polynomial Theory (graduate course)	Spring 2026
Teaching Assistant, Calculus II (undergraduate course)	Spring 2026
Teaching Assistant, Calculus II (undergraduate course)	Spring 2025

Talks

TBD July 2026
Combinatorics Seminar, Sungkyunkwan University.

Details to be announced.

The extended (km, kn) shuffle theorem July 2026
Combinatorics Seminar, Shandong University.

In this talk, I described an extended (km, kn) shuffle theorem for rational parking functions with zero labels. I explained the role of Negut elements on the algebraic side and the extended Delta-type refinement in the (km, kn) setting. This is joint work with Dun Qiu.

The nonsymmetric compositional Delta theorem July 2026
Combinatorics Seminar, Shandong University.

In this talk, I presented a nonsymmetric analogue of the compositional Delta theorem, focusing on flagged LLT polynomial identities and the nonsymmetric versions of the ∇ and τ^* operators. I also discussed how Weyl symmetrization recovers the compositional Delta theorem. This is joint work with Dun Qiu.

Schur positivity of ∇m_μ June 2026
Combinatorics Seminar, Shandong University.

In this talk, I discussed the Schur-positivity problem for ∇m_μ and explained how the nabla operator, the $C_\alpha(1)$ expansion, the compositional shuffle theorem, and LLT positivity enter the proof. This is joint work with Dun Qiu.

Honors and Awards

Gongneng Scholarship, Nankai University	2025
Graduate Recommendation Scholarship, Second Class, Nankai University	2024
Gongneng Scholarship, Nankai University	2024
First Prize, APMCM Asia-Pacific Mathematical Contest in Modeling	2023
Merit Student, Nanjing Normal University	2023
Second Prize, National Undergraduate Mathematics Competition	2022
Second Prize, China Undergraduate Mathematical Contest in Modeling	2022
Honorable Mention, Mathematical Contest in Modeling	2022
Merit Student, Nanjing Normal University	2022
Outstanding Student Scholarship, Nanjing Normal University	2020–2024

Freshman Scholarship, Nanjing Normal University	2020

Miscellanea

Languages:	Mandarin Chinese (native); English.
Computing:	Familiar with LaTeX, Matlab, Maple, SageMath; basic C++, Python, and Julia.

Last updated: July 2026